

DAY 04 — Recon Toolkit Practice

Organization: Trios Cyber
Assessment: Day 04 Assessment
Task: Recon Toolkit Practice
Date: 04 September 2026
Platform: Kali Linux
Target: example.com
Tools: WHOIS, dig, curl, traceroute

1. Executive Summary

This assessment focused on performing basic reconnaissance against the safe public training resource `example.com`.

The assessment used four reconnaissance techniques:

- WHOIS domain information gathering
- DNS record enumeration using `dig`
- HTTP header analysis using `curl`
- Network path observation using `traceroute`

The collected information was preserved in raw log files and supported with terminal screenshots.

No exploitation, credential attacks, brute-force activity, vulnerability exploitation, or destructive activity was performed.

2. Assessment Objectives

The objectives of Day 04 were:

1. Practice WHOIS reconnaissance.
 2. Query DNS records using `dig`.
 3. Analyze HTTP response headers using `curl`.
 4. Observe the network path using `traceroute`.
 5. Identify useful reconnaissance findings.
 6. Create a structured reconnaissance worksheet.
 7. Preserve technical logs and visual evidence.
-

3. Authorization and Scope

The activity was limited to the safe public training resource:

Target: `example.com`

The assessment was restricted to basic reconnaissance techniques.

Out of Scope

The following activities were not performed:

- Exploitation
- Credential attacks
- Brute-force attacks
- Vulnerability exploitation
- Persistence
- Privilege escalation
- Denial-of-service activity

- Destructive actions

4. Lab Environment

Reconnaissance Workstation

- Operating System: Kali Linux
- Purpose: Security reconnaissance and analysis
- Tools: WHOIS, dig, curl, traceroute

Target

- Domain: `example.com`
 - Classification: Safe public training resource
-

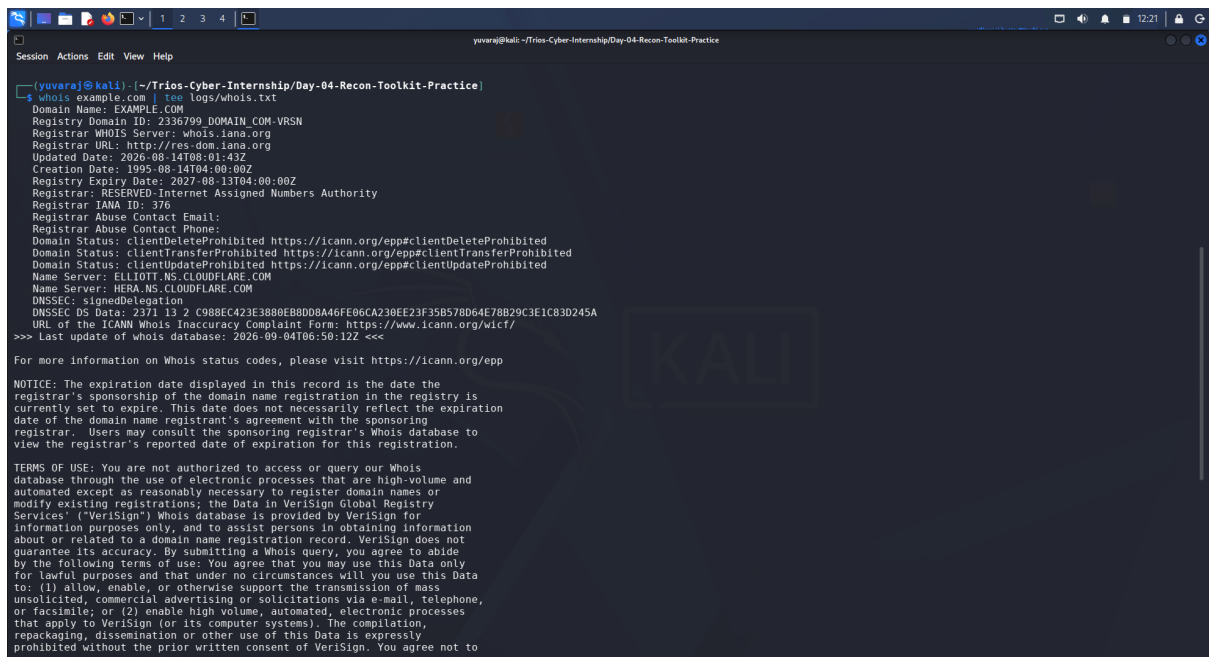
5. WHOIS Reconnaissance

WHOIS was used to collect publicly available domain registration information.

Command

`whois example.com`

Evidence



```
yuvraj@kali: ~/Trio-Cyber-Internship/Day-04-Recon-Toolkit-Practice
$ whois example.com | tee logs/whois.txt
Domain Name: EXAMPLE.COM
Registry Domain ID: 2336799 DOMAIN.COM-VRSN
Registrar WHOIS Server: whois.iana.org
Registrar URL: http://res-dom.iana.org
Updated Date: 2026-08-14T08:01:43Z
Creation Date: 1995-08-14T04:00:00Z
Registry Expiry Date: 2026-08-13T04:00:00Z
Registrar: RESERVED-Internet Assigned Numbers Authority
Registrar IANA ID: 376
Registrar Abuse Contact Email:
Registrar Abuse Contact Phone:
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: ELIOTT.NS.CLOUDFLARE.COM
Name Server: HERA.NS.CLOUDFLARE.COM
DNSSEC: signedDelegation
DNSSEC DS Data: 2371 13 2 C088EC423E3880EB0DD0A46FE06CA230EE23F358578D64E78829C3E1C83D245A
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-09-04T06:50:12Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this Data only
for lawful purposes and that under no circumstances will you use this Data
to: (1) allow, enable, or otherwise support the transmission of mass
unsolicited, commercial advertising or solicitations via e-mail, telephone,
or facsimile; or (2) enable high volume, automated, electronic processes
that apply to VeriSign (or its computer systems). The compilation,
repackaging, dissemination or other use of this Data is expressly
prohibited without the prior written consent of VeriSign. You agree not to
```

Figure 1: WHOIS reconnaissance

Raw Output: logs/whois.txt

The WHOIS output was preserved for later analysis and documentation.

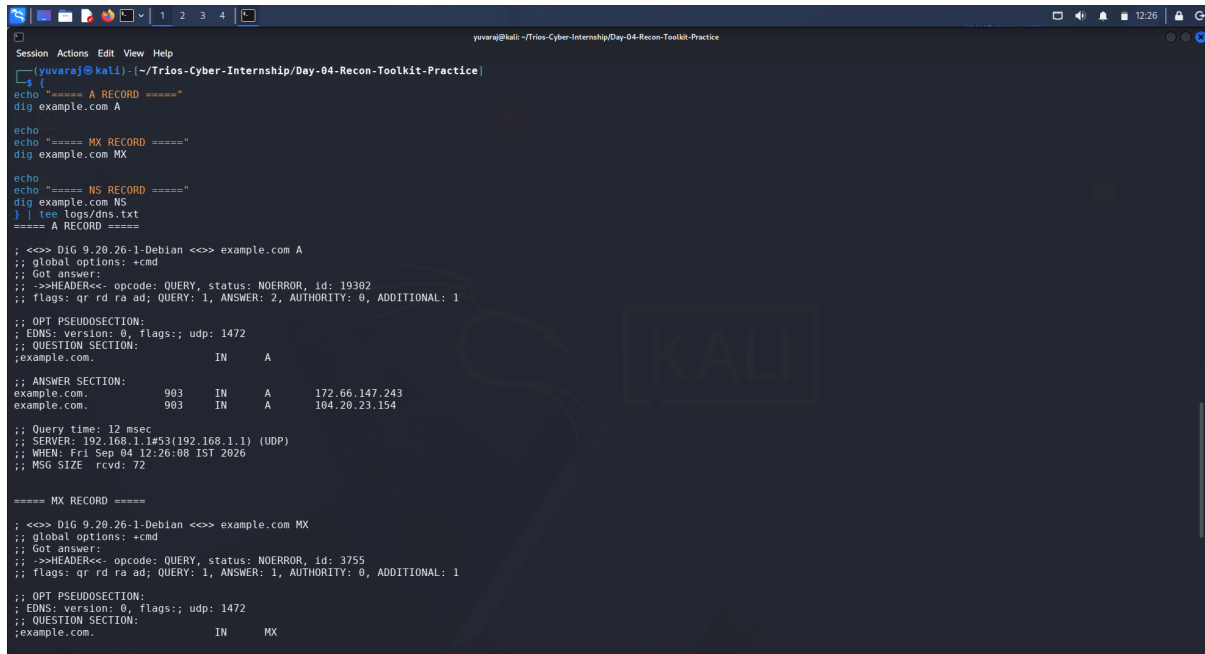
6. DNS Reconnaissance

DNS records were queried using `dig`.

Commands

```
dig example.com A
dig example.com MX
dig example.com NS
```

Evidence



```
yuvraj@kali: ~/Tris-Cyber-Internship/Day-04-Recon-Toolkit-Practice
(yuvraj@kali) ~/Tris-Cyber-Internship/Day-04-Recon-Toolkit-Practice
echo "==== A RECORD ====="
dig example.com A

echo "==== MX RECORD ====="
dig example.com MX

echo "==== NS RECORD ====="
dig example.com NS
} | tee logs/dns.txt
==== A RECORD =====
<<>> Dig 9.20.26-1-Debian <<>> example.com A
;; global options: +cmd
;; Got answer:
;; ->HEADER<<- opcode: QUERY, status: NOERROR, id: 19302
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:;, udp: 1472
;; QUESTION SECTION:
;example.com.
IN      A

;; ANSWER SECTION:
example.com.      903      IN      A      172.66.147.243
example.com.      903      IN      A      104.20.23.154

;; Query time: 12 msec
;; SERVER: 192.168.1.1#53(192.168.1.1) (UDP)
;; WHEN: Fri Sep 04 12:26:08 IST 2020
;; MSG SIZE rcvd: 72

==== MX RECORD =====
<<>> Dig 9.20.26-1-Debian <<>> example.com MX
;; global options: +cmd
;; Got answer:
;; ->HEADER<<- opcode: QUERY, status: NOERROR, id: 3755
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:;, udp: 1472
;; QUESTION SECTION:
;example.com.
IN      MX
```

Figure 2: DNS reconnaissance

Raw Output: logs/dns.txt

The DNS investigation examined:

- A record information
- MX record information
- NS record information

These records provide useful information about domain addressing, mail services, and authoritative name servers.

7. HTTP Header Reconnaissance

HTTP response headers were collected using curl.

Command

```
curl -I https://example.com
```

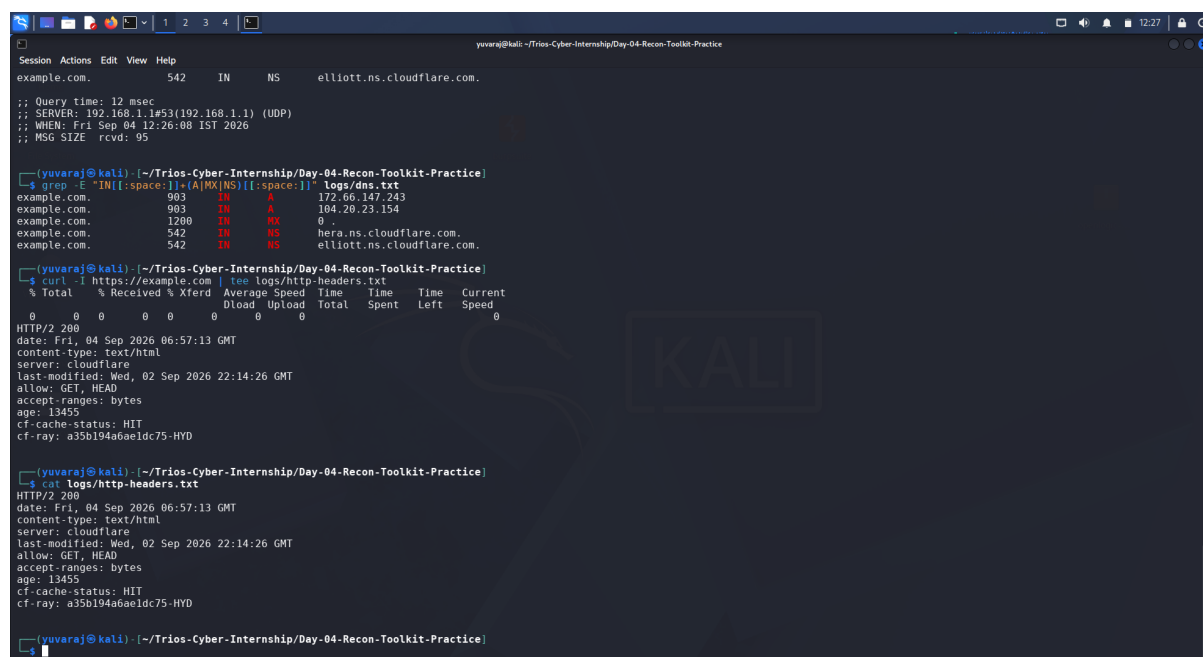
Observed Response

The captured response returned:

- HTTP status: 200
- Content type: text/html
- Server: cloudflare
- Cache status: HIT

Additional response metadata was preserved in the raw log.

Evidence



```
yuvraj@kali: ~/Trio-Cyber-Internship/Day-04-Recon-Toolkit-Practice
Session Actions Edit View Help
example.com. 542 IN NS elriott.ns.cloudflare.com.

;; Query Time: 12 msec
;; SERVER: 192.168.1.1#53(192.168.1.1) (UDP)
;; WHEN: Fri Sep 04 12:26:08 IST 2026
;; MSG SIZE rcvd: 95

[yuvraj@kali: ~/Trio-Cyber-Internship/Day-04-Recon-Toolkit-Practice]
$ grep -E "IN[[:space:]]+(A|MX|NS)[[:space:]]" logs/dns.txt
example.com. 903 IN A 172.66.147.243
example.com. 903 IN A 104.20.23.154
example.com. 1200 IN MX 0 .
example.com. 542 IN NS hera.ns.cloudflare.com.
example.com. 542 IN NS elriott.ns.cloudflare.com.

[yuvraj@kali: ~/Trio-Cyber-Internship/Day-04-Recon-Toolkit-Practice]
$ curl -I https://example.com | tee logs/http-headers.txt
% Total % Received % Xferd Average Speed Time Time Current
% Dload % Upload Total Spent Left Speed
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
HTTP/2 200
date: Fri, 04 Sep 2026 06:57:13 GMT
content-type: text/html
server: cloudflare
last-modified: Wed, 02 Sep 2026 22:14:26 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 13455
cf-cache-status: HIT
cf-ray: a35b194a6ae1dc75-HYD

[yuvraj@kali: ~/Trio-Cyber-Internship/Day-04-Recon-Toolkit-Practice]
$ cat logs/http-headers.txt
HTTP/2 200
date: Fri, 04 Sep 2026 06:57:13 GMT
content-type: text/html
server: cloudflare
last-modified: Wed, 02 Sep 2026 22:14:26 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 13455
cf-cache-status: HIT
cf-ray: a35b194a6ae1dc75-HYD

[yuvraj@kali: ~/Trio-Cyber-Internship/Day-04-Recon-Toolkit-Practice]
$
```

Figure 3: HTTP response headers

Raw Output: logs/http-headers.txt

8. Traceroute Reconnaissance

Traceroute was used to observe the network path toward the target.

Command

```
traceroute example.com
```

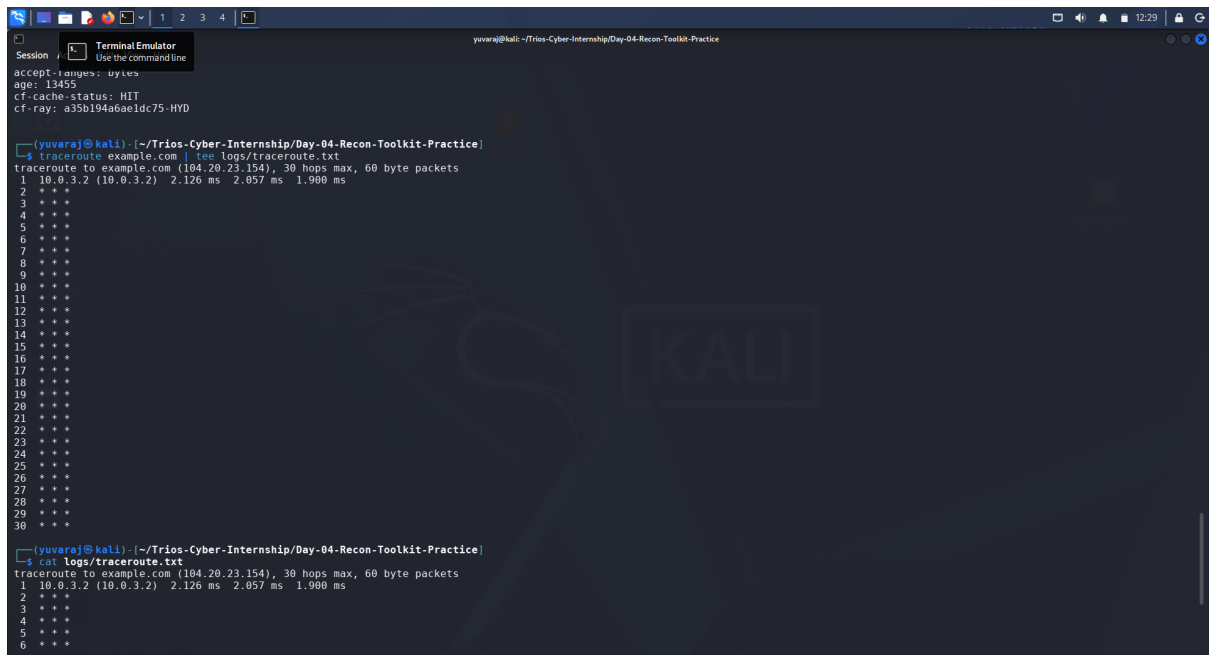
Evidence

Raw Output: logs/traceroute.txt

The traceroute output provides an observation of network hops between the reconnaissance workstation and the destination.

9. Reconnaissance Findings

Category	Tool	Finding
Domain information	WHOIS	Public registration information collected
DNS	dig	A, MX, and NS records queried
HTTP	curl	HTTP 200 response and HTTP metadata observed
Network path	traceroute	Network hops toward the destination observed



```
(yuvraj@kali) [~/Trio-Cyber-Internship/Day-04-Recon-Toolkit-Practice]
$ traceroute example.com | tee logs/traceroute.txt
traceroute to example.com (104.20.23.154), 30 hops max, 60 byte packets
 1  10.0.3.2 (10.0.3.2)  2.126 ms  2.057 ms  1.900 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *

(yuvraj@kali) [~/Trio-Cyber-Internship/Day-04-Recon-Toolkit-Practice]
$ cat logs/traceroute.txt
traceroute to example.com (104.20.23.154), 30 hops max, 60 byte packets
 1  10.0.3.2 (10.0.3.2)  2.126 ms  2.057 ms  1.900 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
```

Figure 4: Traceroute results

10. Final Reconnaissance Summary

The reconnaissance exercise demonstrated how multiple tools can be combined to build an initial understanding of a target's publicly observable infrastructure.

WHOIS provided domain registration information, DNS queries provided records associated with the domain, curl exposed HTTP response metadata, and traceroute provided network-path information.

Evidence

11. Security Observations

Reconnaissance is an important phase of cybersecurity assessment because it helps identify information that may be publicly observable before deeper security testing begins.

Information such as:

- Domain registration details
- DNS records
- IP addressing information
- HTTP server and response metadata
- Network routing information

can contribute to an understanding of an organization's externally visible footprint.

This exercise remained limited to basic reconnaissance and did not attempt to exploit any identified information.

12. Assessment Methodology

The assessment followed this sequence:

1. Define the authorized target and scope.
2. Collect WHOIS information.
3. Query A, MX, and NS DNS records.

```

=====
DAY 04 - RECON TOOLKIT SUMMARY
=====
TARGET: example.com

[1] WHOIS
Domain Name: EXAMPLE.COM
Creation Date: 1995-08-14T04:00:00Z
Registry Expiry Date: 2027-08-13T04:00:00Z
Registrar: RESERVED-Internet Assigned Numbers Authority
Name Server: ELLIOTT.NS.CLOUDFLARE.COM
Name Server: HERA.NS.CLOUDFLARE.COM

[2] DNS
example.com.      903  IN  A    172.66.147.243
example.com.      903  IN  A    104.20.23.154
example.com.     1200  IN  NS   0
example.com.      542  IN  NS   hera.ns.cloudflare.com.
example.com.      542  IN  NS   elliot.ns.cloudflare.com.

[3] HTTP HEADERS
HTTP/2 200
date: Fri, 04 Sep 2026 06:57:13 GMT
content-type: text/html
server: cloudflare
last-modified: Wed, 02 Sep 2026 22:14:26 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 13455
cf-cache-status: HIT
cf-ray: a35b194a6ae1dc75-HYD

[4] TRACEROUTE
21 * * *
22 * * *
23 * * *
24 * * *
25 * * *
26 * * *
27 * * *
28 * * *
29 * * *
30 * * *

=====
RECONNAISSANCE COMPLETE
=====

```

Figure 5: Final reconnaissance summary

4. Collect HTTP response headers.
5. Perform traceroute.
6. Preserve raw command output.
7. Capture visual evidence.
8. Consolidate findings into a reconnaissance worksheet.
9. Document the assessment in a professional report.

13. Evidence Index

Evidence	Description
logs/assessment-scope.txt	Assessment scope and authorization information
logs/whois.txt	WHOIS reconnaissance output
logs/dns.txt	DNS query output
logs/http-headers.txt	HTTP response headers
logs/traceroute.txt	Traceroute output
screenshots/01-whois-recon.png	WHOIS reconnaissance evidence
screenshots/02-dns-recon.png	DNS reconnaissance evidence
screenshots/03-curl-http-headers.png	HTTP header evidence
screenshots/04-traceroute-recon.png	Traceroute evidence
screenshots/05-recon-summary.png	Final reconnaissance summary

14. Learning Outcomes

By completing Day 04, the following concepts were practiced:

- Understanding the purpose of WHOIS.
- Querying DNS records with **dig**.
- Analyzing HTTP response headers with **curl**.
- Observing network paths using **traceroute**.
- Preserving raw reconnaissance output.

- Organizing findings into a structured worksheet.
 - Maintaining an evidence-based cybersecurity assessment workflow.
-

15. Conclusion

Day 04 successfully demonstrated a basic reconnaissance workflow using WHOIS, `dig`, `curl`, and `tracert`.

The exercise showed how different reconnaissance tools provide different categories of information and how those findings can be combined into a structured security assessment.

All collected logs and screenshots were preserved as evidence, and the activity remained within the defined reconnaissance scope.

Assessment Completion Status

Assessment Status: Completed

Scope Status: Basic Reconnaissance Only

Evidence Status: Complete

Worksheet Status: Complete

Report Status: Ready for PDF Generation and Git Submission